
Question 1.1. Let's run a hypothesis test using confidence intervals to see if there is a linear relationship between egg weight and bird weight. Define the null and alternative hypotheses that will allow you to conduct this test. **(8 points)**

Note: Please write your answer **in the cell below** in the following format: - **Null Hypothesis:** - **Alternative Hypothesis:**

Hint: Reminder your null hypothesis should be about the relationship in the population.

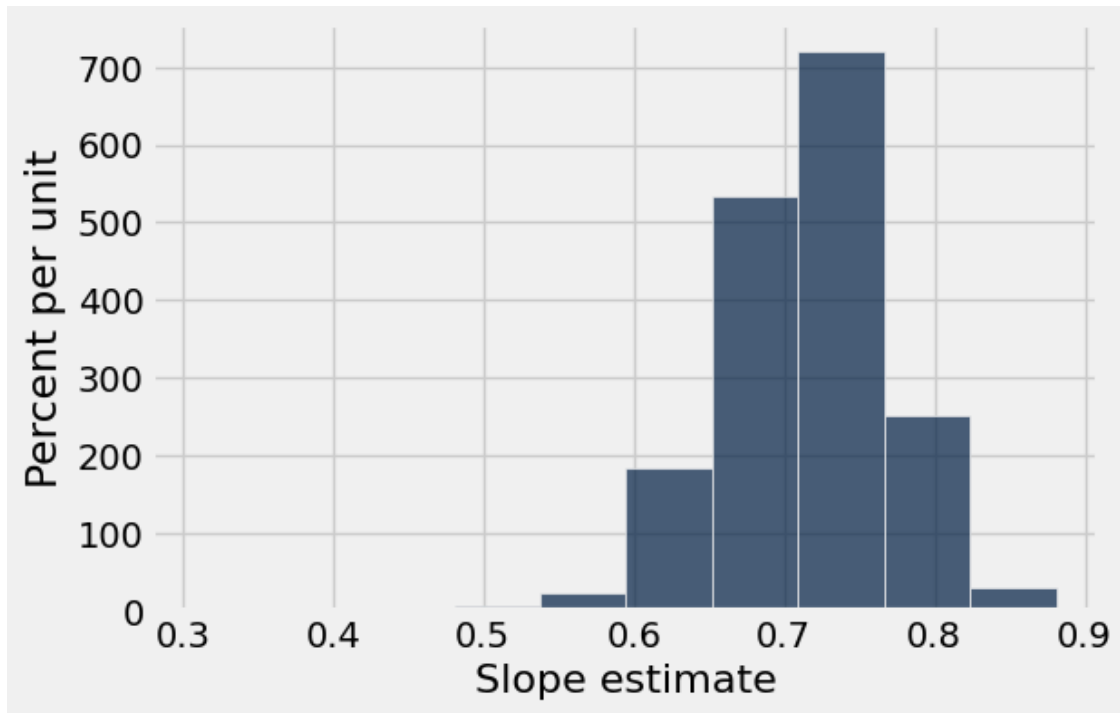
Null Hypothesis: In the population, egg weight and bird weight are not linearly related. (The regression line has a slope of zero)

Alternative Hypothesis: In the population, there is a linear association between egg weight and bird weight. (The regression line's slope is different from zero)

Question 1.4. Create an array called `resampled_slopes` that contains the slope of the best fit line for 1000 bootstrap resamples of `birds`. Plot the distribution of these slopes. (8 points)

```
In [17]: resampled_slopes = make_array()
         for i in np.arange(1000):
             birds_bootstrap = birds.sample()
             bootstrap_line = fit_line(birds_bootstrap, "Egg Weight", "Bird Weight")
             bootstrap_slope = bootstrap_line.item(0)
             resampled_slopes = np.append(resampled_slopes, bootstrap_slope)

         # DO NOT CHANGE THIS LINE
         Table().with_column("Slope estimate", resampled_slopes).hist()
```



```
In [18]: grader.check("q1_4")
```

```
Out[18]: q1_4 results: All test cases passed!
```

Question 1.6. Based on your confidence interval, would you reject or fail to reject the null hypothesis that the true slope is 0? Why? What p-value cutoff are you using? **(8 points)**

Hint: Read the introduction of this homework!

We would reject the null hypothesis that the true slope is 0 because the 95% confidence interval for the slope is $[0.606754, 0.816766]$, which does not include 0. This means there is strong evidence that the slope is significantly different from zero. We are using a significance level (p-value cutoff) of 0.05 to make this decision.

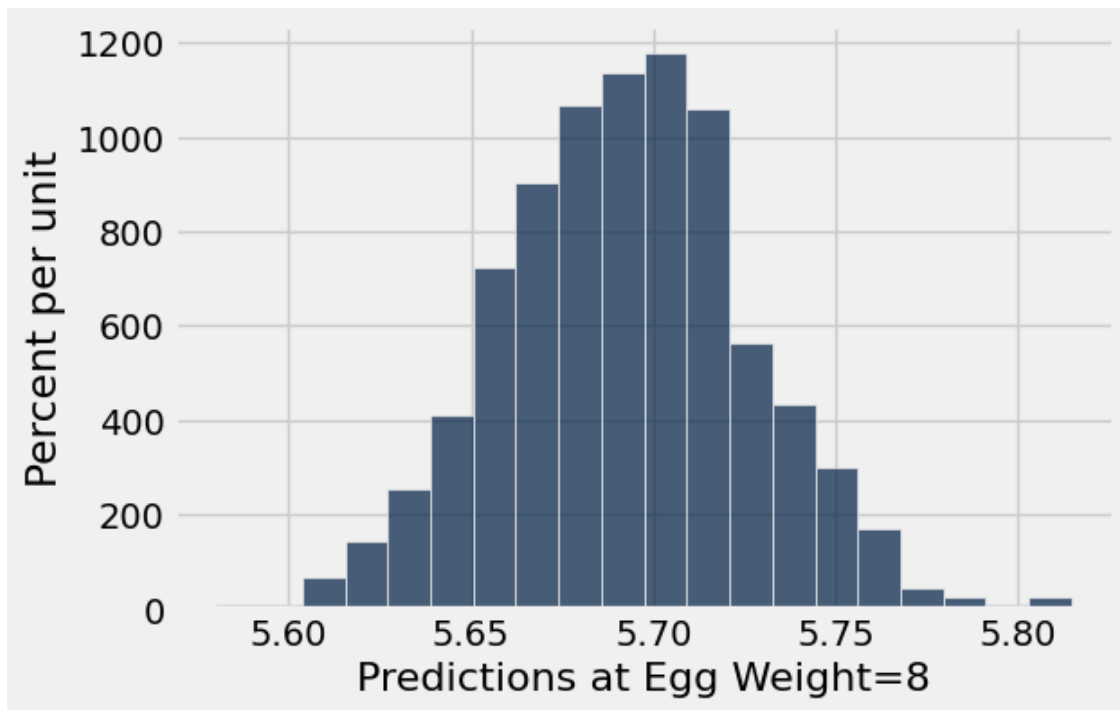
Question 1.7. Using your results from previous questions, **provide an estimated range for the true slope**. How confident are you that the true slope falls within this range? **(8 points)**

Estimated range for the true slope: $[0.606754, 0.816766]$.

Confident that the true slope is in the range: 95%

Question 2.4. Create an array called `predictions_for_eight` that contains the predicted bird weights based on an egg of weight 8 grams for each regression line in `regression_lines`. (6 points)

```
In [37]: predictions_for_eight = np.array([8 * slope + intercept for slope, intercept in zip(regression_lines,
# This will make a histogram of your predictions:
table_of_predictions = Table().with_column('Predictions at Egg Weight=8', predictions_for_eight)
table_of_predictions.hist('Predictions at Egg Weight=8', bins=20)
```



```
In [38]: grader.check("q2_4")
```

```
Out[38]: q2_4 results: All test cases passed!
```

